

JUUKA NIEMELA, Finland

WMO: 027910

Lat: 63.233N

Lon: 29.234E

Elev: 115

StdP: 99.95

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.9	(c) -28.5	(d) -35.1	(e) 0.1	(f) -31.8	(g) -31.4	(h) 0.2	(i) -28.5	(j) 6.5	(k) -3.9	(l) 6.0	(m) -3.7	(n) 0.4	(o) 220	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	26.8	18.6	24.7	17.7	22.8	16.6	20.2	24.7	18.9	23.1	17.7	21.5	2.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	22.1	17.2	12.5	20.7	16.0	11.5	19.7	58.5	24.4	54.1	23.1	50.3	21.6	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.9	(b) 5.2	(c) 4.6	(e) -34.4	(f) 28.8	(g) 4.1	(h) 2.4	(i) -37.4	(j) 30.6	(k) -39.8	(l) 32.0	(m) -42.0	(n) 33.3	(o) -45.0	(p) 35.1		
(a) 5.9	(b) 5.2	(c) 4.6	(e) -33.7	(f) 21.4	(g) 4.7	(h) 2.1	(i) -37.1	(j) 22.9	(k) -39.9	(l) 24.2	(m) -42.5	(n) 25.3	(o) -46.0	(p) 26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-11.1	-9.0	-5.0	1.4	8.2	12.3	15.6	13.5	8.9	2.7	-1.2	-6.5
	DBStd	10.31	8.46	8.08	5.93	4.11	4.30	3.79	3.43	3.35	3.30	3.75	4.54	7.51
	HDD10.0	3157	656	531	464	257	86	19	3	8	59	227	336	512
	HDD18.3	5782	914	764	723	507	314	184	98	154	283	484	586	770
	CDD10.0	436	0	0	0	1	30	87	175	116	26	1	0	0
	CDD18.3	19	0	0	0	0	1	3	13	3	0	0	0	0
	CDH23.3	334	0	0	0	0	31	67	182	54	0	0	0	0
	CDH26.7	54	0	0	0	0	2	8	34	9	0	0	0	0
Wind	WSAvg	2.2	1.9	2.2	2.5	2.3	2.5	2.3	1.9	1.8	2.0	2.3	2.3	2.2
Precipitation	PrecAvg	737	56	48	42	36	48	74	90	80	65	66	67	64
	PrecMax	923	107	93	89	69	142	157	154	181	144	134	153	105
	PrecMin	620	18	6	7	9	6	22	35	19	15	19	13	25
	PrecStd	87	19	22	17	16	28	29	35	37	29	28	28	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.6	8.9	17.2	26.5	28.0	29.9	27.8	20.5	12.5	9.1	5.1
		MCWB	2.3	2.8	3.9	9.0	15.3	17.9	21.2	20.7	15.8	10.7	8.2	4.6
	2%	DB	1.4	2.6	5.4	13.2	23.6	25.4	27.4	24.6	18.0	10.9	6.8	2.9
		MCWB	0.9	1.6	2.0	7.6	14.0	17.5	19.6	18.0	14.3	9.5	6.1	2.2
	5%	DB	0.5	1.3	3.6	10.8	20.5	23.0	25.3	22.4	16.3	9.6	5.3	1.6
		MCWB	0.2	0.8	1.3	5.8	13.0	16.0	18.9	16.8	13.5	8.5	4.9	1.1
	10%	DB	-0.2	0.6	2.2	8.7	17.7	20.6	23.3	20.5	14.9	8.3	3.9	0.8
		MCWB	-0.5	0.1	0.6	4.6	10.9	14.6	17.7	16.2	12.8	7.4	3.6	0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	3.0	4.3	10.2	16.7	20.3	22.2	21.2	16.3	11.5	8.3	4.8
		MCDB	3.0	4.0	7.5	15.3	23.7	24.7	27.1	26.7	19.0	12.4	9.0	5.2
	2%	WB	1.0	1.7	2.7	8.1	15.0	18.6	20.8	18.9	15.2	10.0	6.2	2.4
		MCDB	1.3	2.3	4.7	12.8	21.2	23.8	25.5	22.8	17.1	10.5	6.7	2.7
	5%	WB	0.2	0.8	1.8	6.1	13.6	16.7	19.7	17.7	14.1	8.7	4.9	1.2
		MCDB	0.4	1.2	3.1	10.1	19.5	21.1	24.0	21.4	15.8	9.3	5.3	1.5
	10%	WB	-0.6	0.2	0.9	4.8	11.7	15.3	18.5	16.7	13.1	7.3	3.6	0.6
		MCDB	-0.3	0.6	2.0	7.6	16.8	19.5	22.5	19.8	14.5	8.2	3.9	0.8
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.9	9.9	10.3	12.2	11.3	11.0	10.5	8.7	5.7	4.0	5.6
		MCDBR	3.7	4.3	8.8	14.3	18.4	15.6	14.8	14.2	11.9	7.6	4.4	3.6
		MCWBR	3.7	3.9	6.3	8.8	9.8	8.3	7.9	8.3	8.5	6.5	4.3	3.5
	5% WB	MCDBR	3.5	4.2	7.5	12.7	16.4	13.4	13.2	12.2	10.0	6.6	4.3	3.3
		MCWBR	3.6	3.9	5.9	8.2	9.3	7.3	7.5	7.4	7.4	6.1	4.2	3.3
Clear-Sky Solar Irradiance	taub	0.225	0.282	0.310	0.348	0.348	0.345	0.371	0.352	0.330	0.309	0.240	0.209	
	taud	1.990	2.108	2.170	2.297	2.452	2.490	2.431	2.470	2.519	2.398	2.079	1.969	
	Ebn at Noon	384	635	777	830	868	878	839	825	775	627	363	193	
	Edh at Noon	32	66	93	103	96	95	98	87	68	51	28	15	
All-Sky Solar Radiation	RadAvg	0.18	0.74	2.22	3.67	4.74	5.27	4.99	3.79	2.09	0.79	0.19	0.08	
	RadStd	0.02	0.17	0.27	0.43	0.53	0.32	0.51	0.43	0.24	0.13	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	+0.53	N/A	N/A	N/A	N/A	N/A	-186	N/A	N/A	N/A

Nomenclature: See separate page